

Payal Mohapatra | Curriculum Vitae

PhD candidate, Computer Engineering – Northwestern University, Illinois 60202

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Webpage  Github  LinkedIn

Research Interests : Machine Learning for Time-series, Audio and Multimodal-sensing applications.

Skills

Deep Learning: Sequence Modeling (time-series and audio), Multi-modal Learning, Self-Supervised Learning, Large Language Models (Llama2, Llama3), Real-time Inference, On-device deployment, Signal Processing

Sensors: Inertial Measurement Units (IMUs), Audio, Speech, Surface Electromyography (sEMG), Contact Microphone, Vision, Photoplethysmography (PPG), Electrocardiogram (ECG), Electroencephalogram (EEG), Electrodermal Activity (EDA)

Programming Languages: Python (Pytorch, Scipy, Pandas, Tensorflow), C++, Shell, Perl, Verilog, System Verilog, Universal Verification Methodology(UVM), $\text{\LaTeX}$

Tools: LabVIEW, Matlab, Spice Simulation & PCB Design, Xilinx Vivado, Khadas VIM3, Jetson TX2

Misc.: Human-subjects research, Multidisciplinary and cross-organizational projects

Employment History

Mitsubishi Electric Research Lab (MERL)

Research Scientist Intern, Boston, USA

June,2025 – Aug, 2025

Meta Reality Labs

Part-time Student Researcher / Research Scientist Intern, Redmond, USA

June,2024 – Dec, 2024

Meta Reality Labs

Part-time Student Researcher / Research Scientist Intern, Redmond, USA

June,2023 – Dec, 2023

Analog Devices Incorporation

Senior Design Engineer / Design Verification Engineer, Bangalore, India

Sept, 2021 – Nov, 2017

Indian Institute of Technology Madras

Research Intern, Chennai, India

May,2014 – June, 2014

Hindustan Aeronautics Limited

Control Systems Intern, Bangalore, India

May,2013 – June, 2013

Education

Northwestern University

PhD, Computer Engineering, Illinois, USA

2021– Winter/Spring 2026

Thesis : Empowering Real-World Sensing: Human-Centered Machine Learning for Imperfect Time Series

Advisor : Dr. Qi Zhu

CGPA : 3.96/4

Indian Institute of Technology Madras

Masters (by research), Electrical Engineering, India

2015–2017

Thesis : An Optical Sensor and Wearable Framework for Wearable Cardiac Monitor

Advisor : Dr. Mohanasankara Sivaprakasam

CGPA : 9.31/10

Madras Institute of Technology, Anna University

Bachelor of Engineering, Electronics & Instrumentation, India

2011–2015

Thesis : Advanced Control Schemes for DC Motors

Advisor : Dr. Manamalli Deivasigamani

CGPA : 9.36/10

Selected Awards and Fellowships

2025: RCTP-Q Science Communication Workshop, Northwestern University.
2024: EECS Rising Stars, Massachusetts Institute of Technology.
2023: Top performer in the ACM Multimedia 2023 Computational Paralinguistics Challenge (ComParE).
2023: Top performer in e-Prevention: Person Identification and Relapse Detection from Continuous Recordings of Biosignals Challenge in ICASSP'23.
2022: 1000 USD travel grant for MobiSys'22.
2022: CRA-WP Career Mentoring Workshop
2021: Best research video award Design Automation Conference Young Fellowship (DAC YF).
2018: Best Paper IEEE WinTechCon Conference.
2017: Winner of Make-in-India Anveshan Design Challenge, Analog Devices Incorporation.
2015-2017: Research Fellowship by Government of India (top 2%).
2011-2015: National Merit Scholarship by Government of India.

Publications

In Conference Proceedings.....

2025: Lixu Wang, Bingqi Shang, Yi Li, **Payal, Mohapatra**, Wei Dong, Xiao Wang, and Qi Zhu. Split adaptation for pre-trained vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
2025: **Payal, Mohapatra**, Yueyuan Sui, Akash Pandey, Stephen Xia, and Qi Zhu. Maestro: Adaptive sparse attention and robust learning for multimodal dynamic time series. In *Under review*, 2025.
2025: **Payal, Mohapatra***, Akash Pandey*, Xiaoyuan Zhang*, and Qi Zhu. Can llms understand unvoiced speech? exploring emg-to-text conversion with llms. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL): Main Conference*, 2025. Accepted to the ACL 2025 Main Conference.
2025: Akash Pandey*, **Payal, Mohapatra***, Wei Chen, Qi Zhu, and Sinan Keten. Timesliver: Interpretable time-series classification using symbolic-linear representation. In *Under review*, 2025.
2024: **Payal, Mohapatra***, Shamika Likhite*, Subrata Biswas, Bashima Islam, and Qi Zhu. Missingness-resilient video-enhanced multimodal disfluency detection. In *Interspeech 2024*, 2024.
2023: **Payal, Mohapatra***, Akash Pandey*, Yueyuan Sui*, and Qi Zhu. Effect of attention and self-supervised speech embeddings on non-semantic speech tasks. In *Proceedings of the 31st ACM International Conference on Multimedia*, MM '23, 2023.
2023: **Payal, Mohapatra***, Akash Pandey*, Sinan Keten, Wei Chen, and Qi Zhu. Person identification with wearable sensing using missing feature encoding and multi-stage modality fusion. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.
2023: **Payal, Mohapatra**, Bashima Islam, Md Tamzeed Islam, Ruochen Jiao, and Qi Zhu. Efficient stuttering event detection using siamese networks. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.
2022: **Payal, Mohapatra**, Akash Pandey, Bashima Islam, and Qi Zhu. Speech disfluency detection with contextual representation and data distillation. In *Proceedings of the 1st ACM International Workshop on Intelligent Acoustic Systems and Applications*, 2022.
2017: **Payal, Mohapatra**, SP Preejith, and Mohanasankar Sivaprakasam. A novel sensor for wrist based optical heart rate monitor. In *2017 IEEE international instrumentation and measurement technology conference (I2MTC)*. IEEE, 2017.

Journal Articles.....

2025: **Mohapatra, Payal**, Lixu Wang, and Qi Zhu. Phase-driven domain generalizable learning for nonstationary time series. *Under Review*, 2025.
2025: **Mohapatra, Payal**, Calvin Murdock, Ishwarya Ananthbhola, Ali Aroudi, Anjali Menon, Buye Xu, and

Morteza Khaleghimeybodi. Behavior-driven estimation of acoustic zones of interest using smartglasses. *Under Review*, 2025.

2024: Mohapatra*, Payal, Vasudev Aravind*, Marisa Bisram, Young-Joong Lee, Hyoyoung Jeong, Katherine Jinkins, Richard Gardner, Jill Streamer, Brent Bowers, Lora Cavuoto, Anthony Banks, Shuai Xu, John Rogers, Jian Cao, Qi Zhu, and Ping Guo. Wearable network for multilevel physical fatigue prediction in manufacturing workers. *PNAS Nexus*, volume 3, 2024.

2018: Mohapatra, Payal, Preejith Sreeletha Premkumar, and Mohanasankar Sivaprakasam. A yellow–orange wavelength-based short-term heart rate variability measurement scheme for wrist-based wearables. *IEEE Transactions on Instrumentation and Measurement*. IEEE, 2018.

Patent.....

Systems and Methods for Identifying Zones of Audio Interest via Motion Sensors

Filed: Jan 2025. *Provisional Patent Application* No. 63/746,543. Assignee: Meta Platforms, Inc.

Inventors: **Payal Mohapatra**, Morteza Khaleghimeybodi, Ali Aroudi, Anjali Menon, Calvin Murdock, Ishwarya Ananthbholu, Buye Xu

Preprint Articles.....

2024: Mohapatra, Payal, Ali Aroudi, Anurag Kumar, and Morteza Khaleghimeybodi. Non-verbal hands-free control for smart glasses using teeth clicks. *arXiv preprint arXiv:2408.11346*, 2024.

Selected Research Experience

Meta Reality Labs, Research Scientist Intern | Part-time Student Researcher.....

Behavior-Driven Estimation of Acoustic Zones of Interest Using Smartglasses June 2024 – Dec 2024

Designed a data-driven deep learning architecture with hierarchical sensor fusion to extract conversation-focused features primarily from head orientation data, enabling the estimation of the number and location of speakers in seated group conversations.

Collaborators:: Dr. Morteza Khaleghimeybodi, Dr. Ali Aroudi, Dr. Calvin Murdock, Dr. Buye Xu, Dr. Anjali Menon

Non-Verbal Discreet Communication Technology for Smart Glasses June 2023 – Dec 2023

Pioneered a novel system for Audio Event Detection (AED) on AR glasses using highly sensitive accelerometers. Developed and deployed lightweight, application-specific convolutional inference algorithms for real-time processing. Demonstrated a fully functional prototype at the Meta internal symposium.

Collaborators:: Dr. Morteza Khaleghimeybodi, Dr. Ali Aroudi, Dr. Anurag Kumar

Northwestern University, PhD Candidate.....

Predictive Models for Human Fatigue and Safety – Operator 4.0 2022–2025

Developed sample-efficient machine learning methods to predict perceived fatigue levels from biophysical and locomotive sensor data collected through user studies. Demonstrated a functional closed-loop prototype on two factory floors with near-real-time data visualization and user feedback. Ongoing efforts include building a practitioner’s guide to manufacturing safety using commercial sensors for ergonomic risk estimation and developing a browsable knowledge repository.

Collaborators:: Northwestern University, Boeing, John Deere, MxD, University at Buffalo

Learning from Heterogeneous Modalities 2025

Developed an efficient and robust deep learning framework to learn from large sets of multimodal and dynamic (occasionally missing) data streams. Further explored application-specific strategies for handling heterogeneity in multimodal learning by investigating approaches to incorporate electromyography (EMG) signals into text-based large language models (LLMs), inspired by successful adaptor-based methods used for integrating audio with text-based LLMs.

Low-Overhead Phase Augmentation for General Time Series Analysis 2024–2025

Exploring the role of phase in generating semantically consistent augmented views of time series data for classification, forecasting, and anomaly detection tasks. Designing low-/non-parametric strategies to enable effective augmentation with minimal computational overhead.

Multi-Modal and Self-Supervised Disfluency Detection

2022–2024

Designed robust architectures and training pipelines for disfluency detection under data scarcity and modality unreliability. Curated and released a public multimodal disfluency dataset, explored self-supervised pretraining on real-world unlabeled speech, and demonstrated strong downstream performance with minimal labeled data.

Collaborators:: Dr. Bashima Islam (*Worcester Polytechnic Institute*), Dr. Md Tamzeed Islam (*Amazon Lab126*), Dr. Qi Zhu (*Northwestern University*)

Addressing Non-Stationarity for Domain Generalization in Time Series Applications

2023

Developed a generalizable machine-learning framework by investigating the relationship between non-stationarity and phase in time series. Demonstrated the effectiveness of the method on several time series classification tasks empirically and theoretically.

Indian Institute of Technology Madras, Research Assistant.....

An Optical Sensor and Wearable Framework for Wearable Cardiac Monitor

2015–2017

Designed and developed a custom optical sensor board for optimal signal quality in users with varying skin tones. Developed a motion-artifact rejection algorithm based on normalized least-mean-squared adaptive filtering for real-time processing and validated it with an extensive user study.

Collaborators:: Preejith SP (*Healthcare Technology Innovation Centre*), Dr. Mohanasankar Sivaprakasam (*Indian Institute of Technology Madras*)

Teaching Assistantship

EE5400: Analog and Digital Circuits

IIT Madras
Winter, 2017

EE5401: Measurements and Instrumentation

IIT Madras
Fall, 2016

EE3006: Principles of Measurement

IIT Madras
Winter, 2016

Services

Primary Reviewer: ICASSP'25, Neurips'24, ICLR'24, IEEE IoT Journal, IMWUT'24, IROS'24, ICASSP'24

Secondary Reviewing: ASP-DAC'24, EMSOFT'23, ICCPS'23, NSys'22

Technical Book Reviewing: Why Does Math Work. . . If It's Not Real?, Cambridge University Press'23

Journal Club: NICO Reading Group, Northwestern University, CPS Study Group

Mentoring

Current.....

Bingqi Shang (Masters, Northwestern University)

Xiaoyuan Zhang (Masters, Northwestern University)

Talia-Ben Naim (Masters, Northwestern University)

Mark Zhang (Masters, Northwestern University)

Brooks Hu (Undergrad, Northwestern University)

Past.....

Kiva Joseph (Masters, Northwestern University)

Shamika Likhite (Masters, Northwestern University)

Yueyuan Sui (Masters, Northwestern University)

Devashri Naik (Masters, Northwestern University)

Jinjin Cai (Masters, Northwestern University)

Shangke Liu (Masters, Northwestern University)

Yuqi Ma (Masters, Northwestern University)

Jonathan Li Chen, Ben Forbes, Justin Lau (Undergraduate students, Mechanical Engineering (Mentored for a course project on sensor data analysis for injury detection), Northwestern University)

References

Dr. Qi Zhu: Professor of Electrical and Computer Engineering, Northwestern University

Dr. Aggelos Katssagelos: Professor of Electrical and Computer Engineering, Northwestern University

Dr. Ping Guo: Associate Professor of Mechanical Engineering, Northwestern University

Dr. Bashima Islam: Assistant Professor of Electrical and Computer Engineering, Worcester Polytechnic Institute